

B3S5 Virology

Tutorial 1

I- Historical overview

1- Redi's experiment :

The theory of spontaneous generation was prevalent until the Middle ages. According to this theory, living organisms, such as flies, maggots, microbes, and even rodents, can, under some circumstances, spontaneously develop from non-living matter. For example, maggots (fly larvae) can develop from rotting meat. In 1668, Redi performed an experiment that challenged this theory.

He placed a fresh piece of meat in each of 3 jars. Jar #1 was left open, jar #2 was tightly sealed and jar #3 was covered with a porous gauze. The experiment is illustrated in Figure 1:

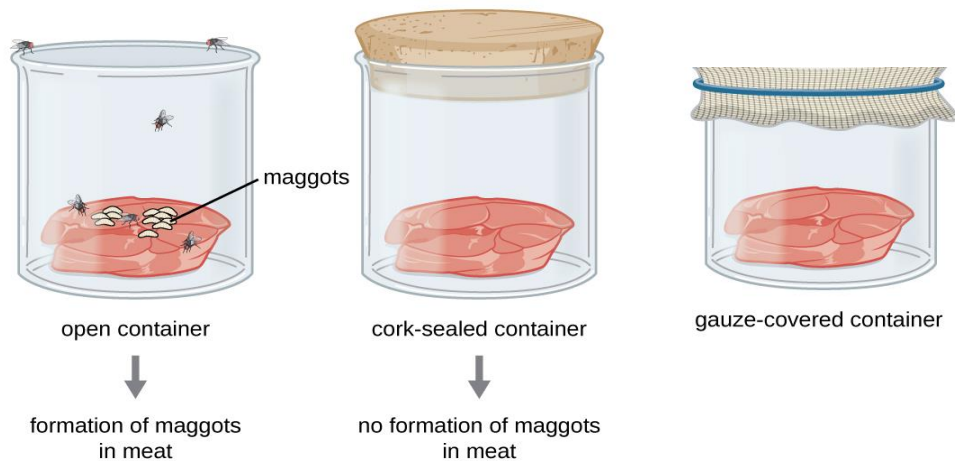


Figure 1: Redi's experiment

- a- What did he observe after a few days? Where there any maggots? If any, where? Where did they come from?
- b- What does this experiment prove? What is the name of the new theory advocated by Redi ?

2- Pasteur's Swan neck experiment

It is very hard to challenge people's beliefs! Despite Redi's work, many scientists still believed in the theory of spontaneous generation until the middle of the 1800's.

Pasteur conducted the experiment of the "Swan neck flask" to further support Redi's theory. This experiment is illustrated in figure 2

A)



B)

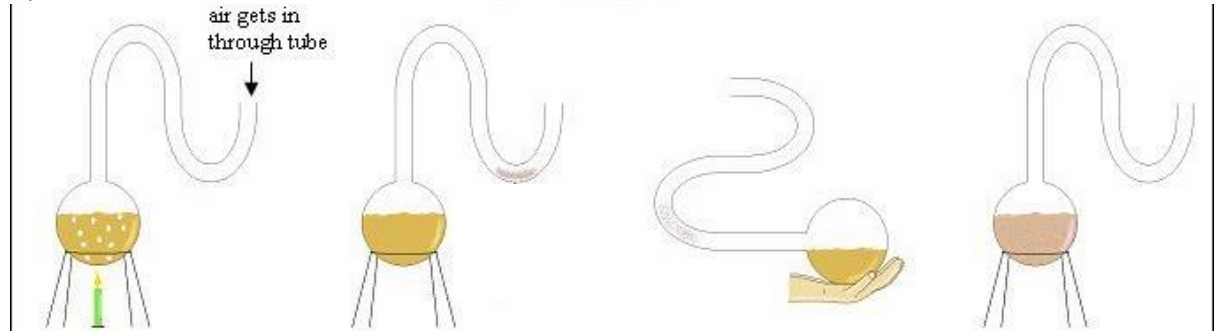


Figure 2: The Swan neck experiment

A) if a flask containing boiled clear broth is left open on a shelf, after several days, it becomes turbid because of the growth of bacteria

B) - If the clear broth is introduced in a "swan neck container" with a curved opening, it can remain clear for a long period of time (months). However, if it is tipped as shown and then returned to a straight position and allowed to sit for several days, it will become turbid.

Interpret this experiment and explain how it further confirmed Redi's theory.

3- Pasteur's chicken cholera experiment

Pasteur observed that chicken cholera could be transmitted through feces from one chicken group to another distant group. He cultured the fecal extract (containing cholera bacterium) in broth, then performed the experiment illustrated in figure 2.

He injected 3 different cholera bacterium cultures to chicken:

- Group 1 : a fresh culture is injected to healthy chicken
- Group 2: an aged culture (that was allowed to sit for several days on a shelf) is injected to healthy chicken
- Group 3: After several days, some chicken from group 2 receive an injection of a fresh culture

Questions:

- What happens to the culture when it is aged for several days?
- What happens to each group of chicken and why?
- How do you interpret this experiment?
- This experiment was a turning point in the history of microbiology and immunology and was further developed in a life-saving medical practice. What is it?

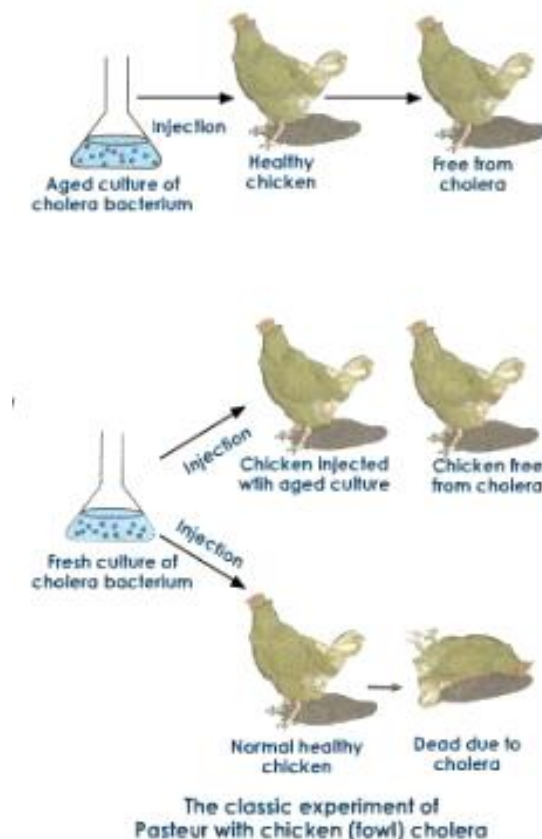
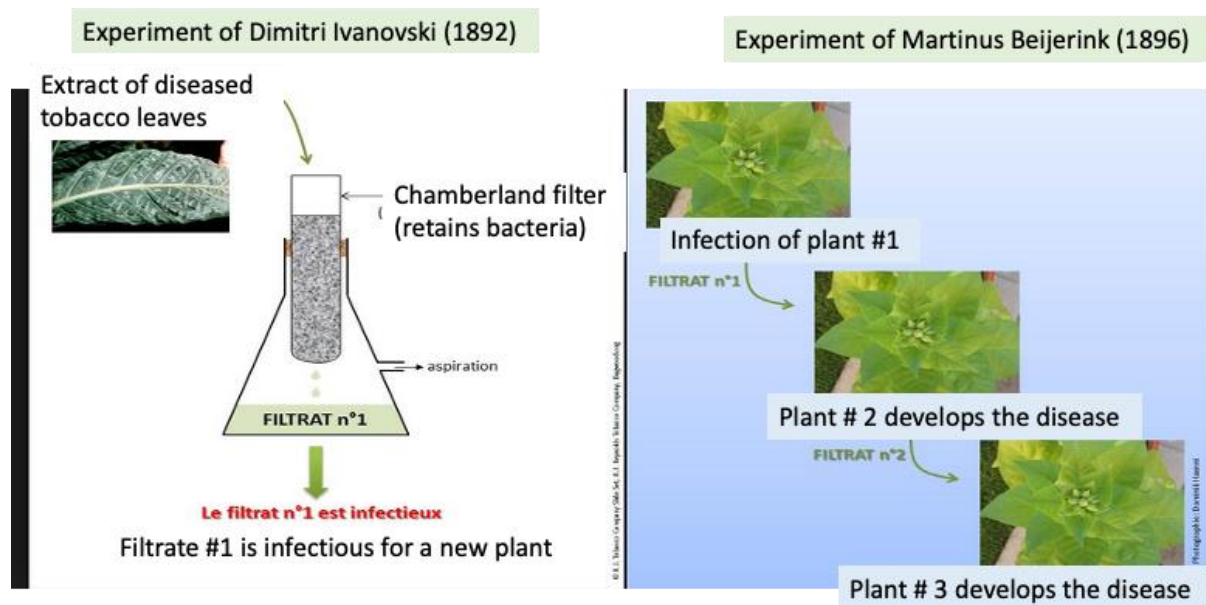


Figure 2

4- Ivanovski's experiment on the Tobacco Mosaic Virus (TMV)

Following the successful experiments of Louis Pasteur proving the bacterial origin of many transmissible diseases, including chicken cholera and rabies, one of his students, Chamberland, designed a porcelain filter with tiny pores that can retain bacteria. When bacteria-contaminated fluids are passed through this filter, they lose their infectivity.

The Tobacco Mosaic disease attacks the leaves of the tobacco plant, inducing yellowing and spotting. To test its microbial origin, Ivanovski performed an experiment, that was later completed by another researcher, Beijerinck. These experiments are illustrated in figure 3:



Questions:

- How do you interpret these experiments?
- Why did all 3 groups of plants in Beijerinck experiment develop the disease?

5- Koch's postulates

Robert Koch was a famous scientist of the XIXth Century who established the bacterial etiology of anthrax and tuberculosis. He formulated four postulates that are used to establish the etiologic relationship between any identified microbe and a given disease.

According to these postulates, the identified organism must:

- 1- Be present in all cases of the disease and absent in all individuals who do not present that disease
- 2- Be isolated from diseased patients
- 3- Cause disease when transmitted to a healthy individual (natural infection) or inoculated in an experimental animal
- 4- Then be isolated again from the new host

Questions:

- Later on, Koch himself rejected the first postulate. Can you think of a given infection situation where this postulate cannot be applied?
- For many decades, viral infections presented a technical problem when it came to apply the second postulate. Why?
- The third postulate is not always true. Can you think of a given infection situation where this postulate cannot be applied?

II- Virus structure and biology

6. What is the key criterion that defines living organisms? Are viruses living organisms? Justify your answer
7. Assign the following properties to bacteria only, viruses only, both or none
 - obligate intracellular parasites
 - genome can be made of either RNA or DNA
 - may have an envelope derived from an animal cell membrane
 - may infect human cells
 - may be grown on acellular nutritive media
 - are always larger than 100 nm in diameter
 - are always smaller than 100 nm in diameter
 - have a capsid
 - have a nucleus
8. Based on their structural properties, in your opinion, which of the enveloped or naked viruses are more capable of infecting humans by ingestion of contaminated water or food?

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